

UTM TENSILE REPORT — UTM_Test_20260817_111700_100%infill_Videocapture3

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A₀) · anchor self-calibration

47.2 MPa

UTS

46.5 MPa

Yield σ_y

2.51 GPa

Modulus E

5.7 %

Failure ϵ_f

2309 kJ/m³

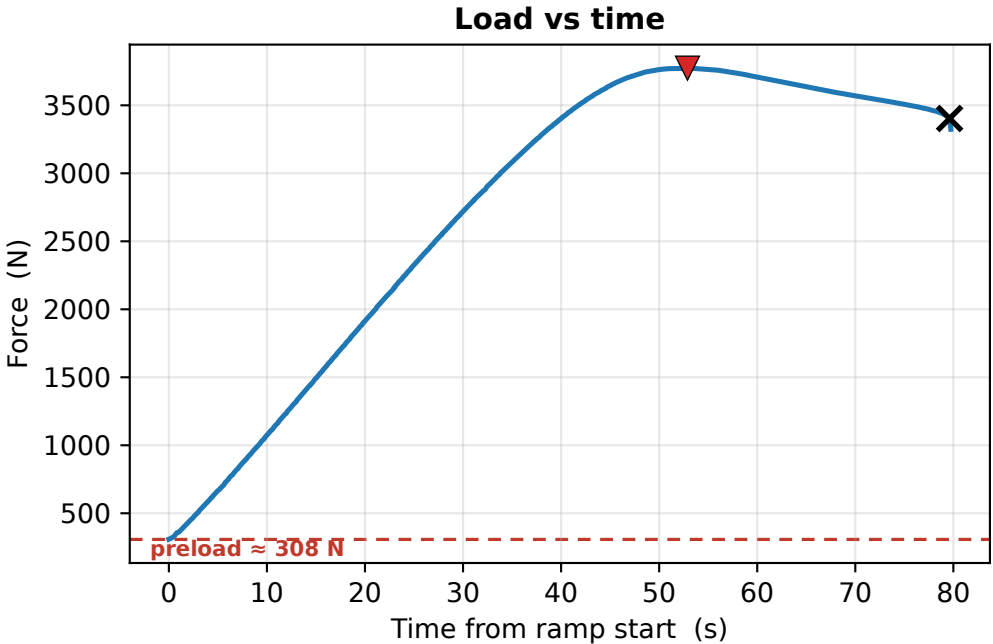
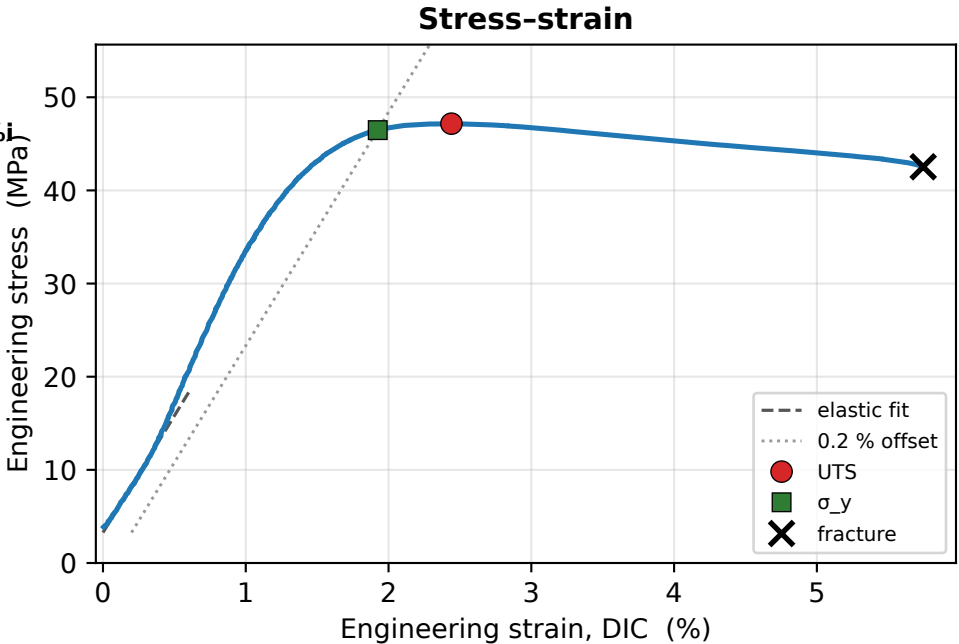
Toughness

308 N

Preload anchor

TEST SETTINGS

Specimen	UTM_Test_20260817_111700_100%infill
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 332.13
Comment	—
Test date	2026-08-17 11:15:15
Duration	125.3 s



VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	47.2	32-60 PASS	×1.23
σ_y	46.5	30-50 PASS	×1.25
E	2.5	3-5.5 low	×1.14
ϵ_f	5.7 %	(range 3-7 %)	×1.39

✓ strength inside journal range (k≈1)

